

# Seungmin Yome

M.S. Student in Energy & Petroleum Engineering

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## Research Experience

### University of Wyoming, Graduate Research Assistant

Laramie, WY, USA  
2025 – present

- Built an axisymmetric finite-volume model (OpenFOAM + solids4Foam) of a spherical proppant grain loaded into and unloaded from Mohr–Coulomb rock through penalty contact; unloading separates permanent embedment from elastic recovery. All rock inputs are published Montney siltstone measurements; nothing is fitted to indentation data
- Compared the model with published Brinell indentation (35 N) and with eight proppant-pack embedment tests predicted blind; the pack comparison shows a grain-size-dependent mismatch that neither mesh refinement nor dilation angle removes, reported as a model limit rather than tuned away
- Showed that residual depth converges slowly with mesh and that the earlier training set carried a mesh bias; now re-running every condition on three mesh levels with Richardson extrapolation and uncertainty bands ( $\approx 750$  runs on UW's MedicineBow cluster, Oct. 2026)
- Fitting a dimensionless residual-depth equation (load, cohesion, friction and dilation angles, grain/rock stiffness) by nested leave-family-out cross-validation; added 60 conditions chosen by a pre-registered balance rule so the equation's range of validity can be stated factor by factor
- Pre-registered the test matrix and pass/fail checks (convergence, force balance) in Git before running, and automated the post-processing so every figure and statistic regenerates from the archived runs
- Trained in triaxial cell operation and DCI rock-compressibility (bulk modulus) testing
- Planned next: inverse models with prediction intervals to find which indentation measurement identifies the dilation angle; preparing the indentation experiments that will supply the data
- Designed and built a displacement-controlled micro-indentation rig for P–h curves on rock and metal (parametric CAD, 3D-printed fixtures, 1 mm tungsten-carbide ball) with a piezoelectric load cell and a laser Doppler vibrometer recorded through an NI DAQ in LabVIEW

### KAIST — Graduate School of Mechanical Engineering, Research Intern (6-month full-time)

South Korea  
2024 – 2025

- Advisor: Dr. Yeunwoo Cho
- Sole investigator on a government-funded study of air-jet-driven waves on a thin elastic film: derived the dispersion relation with film stiffness and tension and confirmed the Kelvin-wave transition on high-speed video
- Assembled a free-surface cavitation tank from parts and explained the upper-wall contraction of ventilated supercavities with a wake-theory term in a 2D model's closure condition, closing most of the gap to the photos
- Simulated von Kármán vortex shedding near a free surface (ANSYS Fluent, VOF): the closer the surface, the lower the Strouhal number and the weaker the upper vortices

**Pusan National University — Department of Naval Architecture & Ocean Engineering,**

Undergraduate Research Assistant

South Korea

2022 – 2024

- Validated RANS CFD of KCS hull resistance against the Gothenburg 2010 benchmark (STAR-CCM+): 1.2% error in total resistance; responsible for mesh refinement at the bow, stern and free surface and for post-processing
- PIV and hull-resistance tests, ITTC 1957/1978 propulsion design, and an ANSYS Fluent study of an airfoil appendage on hull attitude and resistance
- Started the vibration/acoustic anomaly-detection model (MATLAB, Python) finished at Samsung

## Industry Experience

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**Samsung Heavy Industries Laboratory, Mechanical Engineer Intern**

South Korea

2024 – 2024

- Trained an LSTM autoencoder on healthy-condition vibration and acoustic data only (Green Nuri vessel program) so anomalies appear as reconstruction error; detected 621 of 622 test anomalies, one traced to a weld fatigue crack

**Pusan National University, Capstone Design Project**

South Korea

2023 – 2023

- Benchmarked engineering design tools for the shipbuilding industry (AVEVA Marine, TTM, VATIA, AutoCAD) across performance, integration, and usability criteria
- Proposed a cloud-based ML-integrated design platform (Google Cloud) balancing system scalability with market viability

## Publications

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**Development of a dimensionless predictive equation for residual proppant embedment from finite-volume simulations of Montney siltstone**

Jan 2026

Seungmin Yome, Soheil Saraji  
(in preparation)

## Skills

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**Simulation & CAE:** solids4Foam, OpenFOAM, ANSYS Fluent, ABAQUS, STAR-CCM+, ParaView**Programming:** Python, MATLAB, C++, LaTeX**Design & Engineering Tools:** AutoCAD, AVEVA Marine, TTM, VATIA, Google Cloud Platform**Laboratory:** Micro-indentation (rig built in-house), load cell, laser vibrometry, NI DAQ/LabVIEW, PIV, hull resistance testing; triaxial and DCI testing (trained)

## Education

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**University of Wyoming, Energy & Petroleum Engineering**

Laramie, WY, USA

2025 – present

- Advisor: Dr. Soheil Saraji (Department Head)
- Minor: Computer Science (in progress)
- Thesis: Numerical modeling of proppant-supported fracture systems — embedment, rock deformation, and single-phase fluid transport using solids4Foam
- Coursework: Rock Mechanics, Enhanced Oil Recovery, Geostatistics, Python for Engineers
- Current coursework (Fall 2026): ML for Computational Solid Mechanics, Practical Machine Learning

**Pusan National University, Naval Architecture & Ocean Engineering**

South Korea

2018 – 2024

- Advisor: Dr. Inwon Lee
- Coursework: Solid Mechanics, Fluid Mechanics, Linear Algebra, Thermodynamics, Ocean Wave Mechanics, FEM, Offshore Structural Design, CFD

## Grants & Fellowships

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| <b>Graduate Research Assistant, University of Wyoming (fully funded)</b> <ul style="list-style-type: none"><li>University of Wyoming</li></ul>    | 2025–Present |
| <b>Research Scholarship</b> <ul style="list-style-type: none"><li>Korea Advanced Institute of Science and Technology</li></ul>                    | Jan 2024     |
| <b>Undergraduate Research Student Grants</b> <ul style="list-style-type: none"><li>Pusan National University</li></ul>                            | 2022–2024    |
| <b>Academic Excellence Scholarship</b> <ul style="list-style-type: none"><li>Pusan National University</li></ul>                                  | 2022–2024    |
| <b>Foreign Language Special Lecture Consignment Program Scholarship</b> <ul style="list-style-type: none"><li>Pusan National University</li></ul> | Jan 2021     |
| <b>Creative Human Resources Development Project Group Scholarship</b> <ul style="list-style-type: none"><li>Pusan National University</li></ul>   | Jan 2018     |

## Selected Activities & Awards

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| <b>Invited to Samsung Heavy Industries Wind Propulsion Ship Forum (travel award)</b> <ul style="list-style-type: none"><li>Samsung Heavy Industries</li></ul>                                                      | July 2024 |
| <b>Participating in ‘Conceptual Design of Efficient Devices of a 25-tonne High-Speed Ship’</b> <ul style="list-style-type: none"><li>Korean Society of Shipbuilding and Marine Engineering / Coast Guard</li></ul> | Jan. 2024 |
| <b>Invited to Hyundai Heavy Industries Eco-Friendly Smart Ship Forum (travel award)</b> <ul style="list-style-type: none"><li>Hyundai Heavy Industries</li></ul>                                                   | Apr. 2023 |
| <b>TPC (Test Preparation Course) Certification — English Consignment Program</b> <ul style="list-style-type: none"><li>Pusan National University</li></ul>                                                         | July 2021 |

## Teaching & Tutoring

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| <b>Private Tutoring (Educational Volunteering), Mathematics Instructor</b> <ul style="list-style-type: none"><li>Taught high-school mathematics as an educational volunteer for two years</li></ul>                                                                      | South Korea<br>2022 – 2024 |
| <b>Private Tutoring, Physics &amp; Chemistry Instructor — College Entrance Preparation</b> <ul style="list-style-type: none"><li>Taught physics and chemistry to ten high-school seniors preparing for the Korean college entrance exam (CSAT) over six months</li></ul> | South Korea<br>2023 – 2024 |

## Languages

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**Korean:** Native  
**English:** Fluent